FACT: Fact-Augmented Claim Tester for Detecting Hallucination in LLMs

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Introduction

- Large Language Models (LLMs) have become widely adopted for their ability to generate human like text with reasoning capability.
- However, they are prone to hallucinations, making them difficult to trust.

Example:

"Anne Frank and her sister Margot were found alive in the Bergen-Belsen concentration camp after new research was released by the Anne Frank House. Witnesses at the camp stated that they saw the sisters alive as late as February 7, contradicting the date of death previously believed. The sisters endured terrible living conditions and disease, but ultimately survived the Holocaust."

- This project presents a fact-checking pipeline designed to detect hallucinated claims in LLM outputs and retrieve transparent evidence to verify these claims.

Background

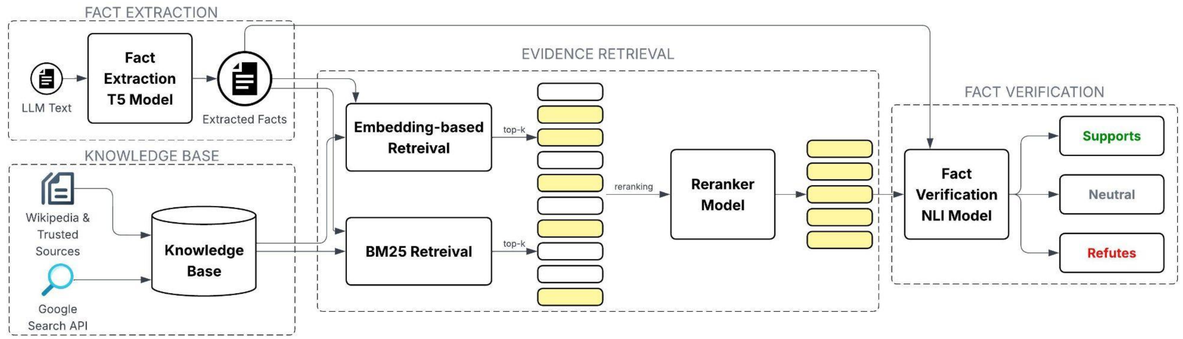
- Research by OpenAI shows that their reasoning models such as OpenAI o3 and o4-mini hallucinate more than their previous models.

Table 4: Hallucination evaluations

Dataset	Metric	o3	o4-mini	o1
SimpleQA	accuracy (higher is better)	0.49	0.20	0.47
	hallucination rate (lower is better)	0.51	0.79	0.44
PersonQA	accuracy (higher is better)	0.59	0.36	0.47
	hallucination rate (lower is better)	0.33	0.48	0.16

- Solutions such as Retrieval-Augmented Generation (RAG) provide information to ground the responses on during generation, but fails to verify the final responses.

System Design



1. Fact Extraction
- The first stage decomposes LLM outputs into atomic claims, where each claim represents a single, verifiable fact, using a fine-tuned Google's Flan-T5 transformer model.

2. Knowledge Base
- The second stage builds the knowledge base by compiling text from trusted truth sources, such as Wikipedia and dynamically updates with Google Custom Search JSON API for niche or recent topics.

3. Evidence Retrieval
- The third stage retrieves supporting evidence from the knowledge base using the two-stage retrieval framework; hybrid retrieval using embeddings and BM25 score retrieves a set of documents that is reranked by a Cross-Encoder model to output a final set of most relevant evidence.

4. Fact Verification
- The fourth stage verifies each extracted fact with the evidence using a DeBERTa-based natural language inference (NLI) model to classify their relationship as **SUPPORTS**, **REFUTES** or **NEUTRAL**.

"Anne Frank and her sister Margot were found alive in the Bergen-Belsen concentration camp after new research was released by the Anne Frank House. Witnesses at the camp stated that they saw the sisters alive as late as February 7, contradicting the date of death previously believed. The sisters endured terrible living conditions and disease, but ultimately survived the Holocaust."

CLAIM: Anne Frank and her sister Margot were found alive in the Bergen-Belsen concentration camp.

Evidence Text: In October or November 1944, Anne and her sister, Margot, were transferred to Bergen-Belsen concentration camp from Auschwitz, where they died - probably of typhus - a few months later. They were originally estimated by the Red Cross to have died in March, with Dutch authorities setting 31 March as their official date of death, but research by the Anne Frank House in 2015 suggests they more likely died in February.

Verification Result: REFUTES (Score: 0.901)

CLAIM: Witnesses at the camp stated that they saw the sisters alive as late as February 7.

Evidence Text: The disease causes high fever, chills and skin eruptions. "Because of the lice infesting the bedstraw and her clothes, Anne was exposed to the main carrier of epidemic typhus for an extended period," museum researchers wrote. They concluded that it's unlikely the sisters survived until March, because witnesses at the camp said the sisters both had symptoms before February 7.

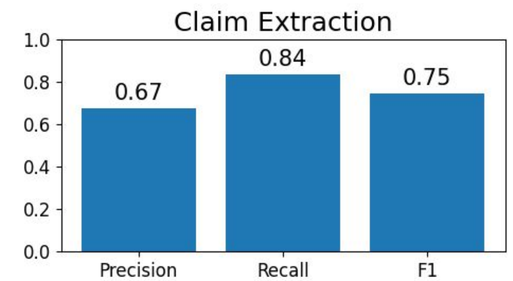
Verification Result: REFUTES (Score: 0.834)

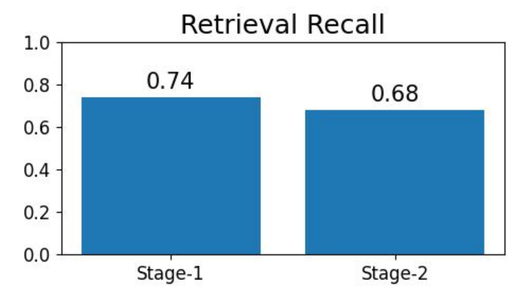
CLAIM: The sisters ultimately survived the Holocaust.

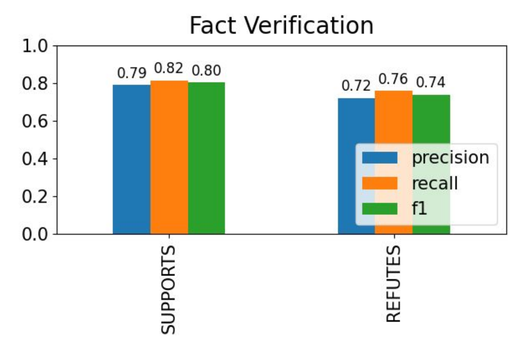
Evidence Text: Seventy years ago, Anne Frank died of typhus in a Nazi concentration camp at the age of 15. Just two weeks after her supposed death on March 31, 1945, the Bergen-Belsen concentration camp where she had been imprisoned was liberated -- timing that showed how close the Jewish diarist had been to surviving the Holocaust. But new research released by the Anne Frank House shows that Anne and her older sister, Margot Frank, died at least a month earlier than previously thought.

Verification Result: REFUTES (Score: 0.987)

Results







Future Work

- more extensive evidence retrieval such as multi-hop retrieval
- improved claim extraction methods to better handle more complex responses.

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